

MARY ANNE SMART

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Assistant Teaching Professor
Department of Computer Science
Purdue University

EDUCATION

University of California San Diego Doctor of Philosophy in Computer Science Dissertation Title: Understanding and Improving Privacy Communication Advisor: Kristen Vaccaro	2024
University of California San Diego Master of Science in Computer Science	2021
Indiana University Bachelor of Science in Computer Science, Bachelor of Science in Math, Bachelor of Arts in Spanish	2017

PROFESSIONAL APPOINTMENTS

Purdue University - Assistant Teaching Professor in Department of Computer Science	2024 - Present
University of California San Diego - Graduate Research Assistant / Instructional Assistant	2018 - 2024
Columbia University - Visiting Ph.D. Student	2022
Indiana University - Undergraduate Research Assistant	2017
Qualtrics - Engineering Services Fullstack Software Developer	2017 - 2018
Proofpoint - Software Engineering Intern	2016
Indiana University - Peer-Led Team Learning Leader	2014 - 2016

TEACHING EXPERIENCE

Purdue University	
· CS 243: Artificial Intelligence Basics (Indianapolis) Enrollment: 9 students	Fall 2025
· CS 290: Usable S&P (Indianapolis) Enrollment: 10 students	Fall 2025
· CS 182: Foundations of Computer Science (Indianapolis) Enrollment: 34 students	Fall 2025
· CS 182: Foundations of Computer Science (Indianapolis) Enrollment: 121 students	Spring 2025
· CS 182: Foundations of Computer Science (West Lafayette) Enrollment: 275 students	Fall 2024
University of California San Diego	
· Intro to Artificial Intelligence: Probabilistic Reasoning (Associate Instructor)	Summer 2023
· Critical Analysis in Computing (TA)	Spring 2023
· Software Tools & Techniques Lab (TA)	Winter 2023
· Human-AI Interaction (TA)	Fall 2021, Fall 2022
Indiana University	
· Introduction to Computer Science (Peer-Led Team Learning Leader)	Fall 2014

- Discrete Structures for Computer Science (Peer-Led Team Learning Leader)
- Mathematical Foundations of Informatics (Peer-Led Team Learning Leader)

Fall 2015
Spring 2015, Spring 2016

RESEARCH INTERESTS

Privacy & Security, Human-Computer Interaction, Education

PUBLICATIONS

Peer-Reviewed Conference / Journal Papers

- **Mary Anne Smart**, Priyanka Nanayakkara, Rachel Cummings, Gabriel Kaptchuk, and Elissa M. Redmiles. Models Matter: Setting Accurate Privacy Expectations for Local and Central Differential Privacy. *Proceedings on Privacy Enhancing Technologies Symposium (PoPETS)*, pp. 653-678, 2025.
- Priyanka Nanayakkara, **Mary Anne Smart**, Rachel Cummings, Gabriel Kaptchuk, and Elissa M. Redmiles. What Are the Chances? Explaining the Epsilon Parameter in Differential Privacy. In *32nd USENIX Security Symposium (USENIX Security 23)*, pp. 1613-1630. 2023.
- **Mary Anne Smart**, Dhruv Sood, and Kristen Vaccaro. Understanding Risks of Privacy Theater in Differential Privacy. *Proceedings of the ACM on Human-Computer Interaction 6*, pp. 1-24. Computer-Supported Cooperative Work and Social Computing (CSCW), 2022.
- Zachary Izzo, **Mary Anne Smart**, Kamalika Chaudhuri, and James Zou. Approximate Data Deletion from Machine Learning Models: Algorithms and Evaluation. In *International Conference on Artificial Intelligence and Statistics*, pp. 2008-2016. Proceedings of Machine Learning Research (PMLR), 2021.

Workshop Papers / Posters

- Matthew Galej and **Mary Anne Smart**. How Students Evaluate the Security of AI-Generated Code. Privacy Enhancing Technologies Symposium (PETS), 2025.
- Tianyi Shan, **Mary Anne Smart**, Jay Jhaveri, Jessalyn Wang, and Kristen Vaccaro. Misalignments Between User and Developer Privacy Threat Models. Workshop on Privacy Threat Modeling at the Symposium on Usable Privacy and Security (SOUPS), 2023.
- **Mary Anne Smart**, Priyanka Nanayakkara, Rachel Cummings, Gabriel Kaptchuk, and Elissa M. Redmiles. Improving Explanations to End Users About Differential Privacy. Privacy Interventions and Education (PIE) Workshop at the Conference on Human Factors in Computing Systems (CHI), 2023.
- Priyanka Nanayakkara, **Mary Anne Smart**, Rachel Cummings, Gabriel Kaptchuk, and Elissa M. Redmiles. Improving Communication with End Users About Differential Privacy. Symposium on Applications of Contextual Integrity, 2022.
- **Mary Anne Smart** and Daniel Tan. Privacy and Security Challenges in Doctoral Students' Human Subjects Research. Symposium on Usable Privacy and Security (SOUPS), 2022.
- **Mary Anne Smart**. Experimental Design Considerations for Limiting Psychological Distress. Queer in AI Workshop at the International Conference on Machine Learning (ICML), 2022.
- Priyanka Nanayakkara, **Mary Anne Smart**, Rachel Cummings, Gabriel Kaptchuk, and Elissa M. Redmiles. Improving Communication of DP to End Users. Theory and Practice of Differential Privacy (TPDP) Workshop at the International Conference on Machine Learning (ICML), 2022.
- **Mary Anne Smart**. Toward Collective Agency in Privacy Management. Workshop on Interdependent and Multi-party Privacy at the Privacy Enhancing Technologies Symposium (PETS), 2022.
- **Mary Anne Smart**. Addressing Privacy Threats from Machine Learning. Human-Centered AI (HCAI) Workshop at the Conference on Neural Information Processing Systems (NeurIPS), 2021.

INVITED TALKS

“Introduction to Linked Lists” Guest lecture at University of Illinois Chicago (2023)

“Lessons Learned in Communicating Differential Privacy to Data Subjects” Lightning talk at Symposium on Usable Privacy and Security (2023)

“Conceptualizations of Privacy in CS Syllabi.” Lightning talk at Consortium for Computing Sciences in Colleges (CCSC) Southwest Region Conference (2023)

“Human-Computer Interaction Research.” Invited talk for High School Women in STEM Club (2023)

“Understanding and Improving Privacy Communication.” Speaker for UC San Diego’s HCI Seminar (2023)

“Differential Privacy in Practice.” Guest lecture at Northwestern University, co-taught with Priyanka Nanayakkara (Winter 2023)

“Differential Privacy in Practice.” Guest lecture at Northwestern University, co-taught with Priyanka Nanayakkara (Fall 2022)

“Limitations of Consent for Ethical Data Collection.” Speaker at Data Ethics and Justice Workshop for UC Love Data Week (2022)

“Data Feminism Discussion: From Understanding Algorithms to Understanding Context.” Invited to lead a discussion with the Socially Responsible Tritons group at UCSD (2020)

ORGANIZED EVENTS

Co-led “Improving AI Education” breakout discussion session at the 2022 WiML Un-Workshop

Co-led “Data Feminism and Machine Learning” breakout discussion at the 2020 WiML Un-Workshop

Co-organized Privacy Preserving Machine Learning Workshop at the 2020 Conference on Neural Information Processing Systems (NeurIPS)

Co-organized Privacy in Machine Learning Workshop at the 2019 Conference on Neural Information Processing Systems (NeurIPS)

MENTORSHIP

Purdue Students 2024 - Present

- Matthew Galej
- Cheyenne Huggins

Horizons Faculty Mentorship Program 2024

- Met with Matthew Galej (undergraduate CS student) three times for “Glow and Grow sessions” to discuss career goals, stress management, academic success, etc.
- Participated in various mentorship program events with first-generation and/or low-income Purdue students

UC San Diego Students 2020 - 2024

- Mentored and collaborated with undergraduate and junior graduate students on research projects, including: Dzhangir Bayandarov, Aaron Broukhim, Emily Jones, Stephanie Li, Tyus Liu, Baljot Sekhon, Dhruv Sood, & Stephen Tan
- Published “Understanding Risks of Privacy Theater in Differential Privacy” in the Conference on Computer-Supported Cooperative Work and Social Computing (CSCW) with undergraduate Dhruv Sood as a co-author

Graduate Women in Computing Mentorship Program 2019 - 2024

- Participated as a student mentor
- Met regularly with students to discuss classes, research, etc.

Explore Computer Science Research Program 2020

- Mentored a small group of local community college students
- Help students learn about nearest neighbor classification, Jupyter notebooks, and data processing in Python
- Discussed the “coded gaze” and related issues that arise when building and deploying AI systems

INSTITUTIONAL SERVICE

Presentation Reviewer, Purdue Undergraduate Research Conference Spring 2025

CS Undergraduate Study Committee, Purdue University Fall 2024, Spring 2025

PROFESSIONAL SERVICE

Reviewer for IUI (2025)
Reviewer for CHI (2024)
Reviewer for CSCW (2024)
Reviewer for CHI Late-Breaking Work (2023)
Volunteer for Black in AI and Queer in AI joint social at ICML (2022)
Reviewer for WIPS (2022)
Reviewer for DIS (2022)
Reviewer for ICWSM (2022)
Reviewer & Volunteer for NeurIPS Queer in AI Workshop (2021)
Reviewer for CSCW Posters (2021)
Reviewer for NeurIPS Resistance AI Workshop (2020)
Reviewer NeurIPS Retrospective Workshop (2020)
Volunteer for GLBIO (2017)

HONORS & AWARDS

Rising Stars in EECS (2023): selected to participate in a workshop for scholars from historically marginalized genders who are interested in academic careers

Doctoral Award for Excellence in Contributions to Diversity (2023): awarded annually to one doctoral student by UCSD's CSE department

Cultural Competence in Computing (3C) Fellow (2021): professional development program hosted by Duke University

Phi Beta Kappa Member (2017): academic honor society

SWSIS Scholar (2016): national scholarship awarded by the Scholarship for Women Studying Information Security program on behalf of ACSA, CRA-W, and HPE

Ruth E. Gilliat Memorial Scholarship (2016): awarded by IU to a student in mathematics who has performed at the highest level in a rigorous curriculum

Aruna & Anurag Mendhekar Scholarship (2016): awarded by IU's School of Informatics and Computing

PROFESSIONAL DEVELOPMENT & TRAINING

IMPACT Week (2025)
Faculty Success Program (NCFDD) (2025)
Certificate of Foundations in College Teaching (2025)
QPR Suicide Prevention (2024)